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Application of rough set and netnography in tourism marketing analysis

Application of
rough set and
netnography

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Abstract

Purpose – Social media has given customers more power over sharing their knowledge, opinions and experiences with each other. Tourists as customers of destinations are also using text ads on social media and websites to share their experiences. The purpose of this study is to find out the factors which have affect on the decisions of tourists towards the most popular destinations in Tehran, Isfahan and Shiraz of Iran.

Design/methodology/approach – Netnography methodology has been applied to 2,852 comments showing travelers' experiences through TripAdvisor.com. As a result, ten major factors have been discovered. According to these factors, a questionnaire has been designed and distributed among 449 tourists. In the second step, the collected data are used by rough set theory to discover the rules of destination recommendation based on the factors discovered before. Finally, eight main rules are determined to further analysis.

Findings – The findings confirm that beauty, cultural attractions, safety, welfare, costs and dealing with passengers are more important than other observed dimensions.

Originality/value – In this study, first the factors affecting consumer behavior in the tourism industry have been investigated. Based on this, the comments of tourists who have traveled to one of the cities Shiraz, Isfahan or Teheran and shared their experiences on TripAdvisor.com are studied. Further, the rules are discovered based on the rough set theory, and owing to the large number of objects (449 customer), the Rosetta software has been used.

Keywords Management information systems, Social networks, Decision analysis, Traveling

Paper type Research paper

1. Introduction

Recently, online-based social networks have been becoming popular in tourism management because the consumers and corporations have a scope to interact with others before selection a destiny. The consumers are habituated in search for useful information from people with a high online status in an online social network (Liu *et al.*, 2015; Lu *et al.*, 2012). Generally, at the time of selection a product or service, the people hunt for recommendations from the users. They prefer generally personal recommendation of quality of services from their friends or trusted acquaintances



(Mochalova and Nanopoulos, 2014). This type of advertising is used widely in websites and social media, and it usually works through peoples' comment.

A growing service that can benefit from the advantages of word-of-mouth (WOM) advertising is tourism industry. It is an important source of income which influences cultural exchange among countries. Certainly, there would be a clear future for this industry. Successful achievements in this industry should also be studied according to this industry's role in improving quality of life, developing social relations and recognizing nations and various beliefs (Taleghani *et al.*, 2014). This industry has a potentiality to use WOM strategy in network as an advertising strategy which has an advantage of the effect of "word-of-mouth" among the relationships of individuals to encourage a product instead of a traditional broadcasting to a massive number of users (Long and Wong, 2014).

The proposed study intends to evaluate and identify different stages in influencing user comments and transfer of WOM by detecting the factors influencing tourist's behavior and using them to discover the rules governing on tourist's behavior toward destinations using rough set theory. In other words, this study proposes a framework to detect factors influencing customers' choices and then extracts the rules of destination marketing based on these factors. This framework helps tourism managers discover the factors from deep layers of people's language on social media and then finds the rules defining relations between these factors. Then based on this information, managers could act as good decision-makers.

To detect factors influencing customer behavior, we use a netnographic approach. "Netnography," or ethnography on the internet, is a new qualitative research methodology that adapts ethnographic research techniques to study the cultures and communities that are emerging through computer-mediated communications (Kozinets, 2002). As Kozinets (2002) stated, netnography is capable of being conducted in a manner that is entirely unobtrusive. Compared with focus groups and personal interviews, netnography is far less obtrusive, because it is conducted using observations of consumers in a context that is not fabricated by the marketing researcher. It can also provide information in a manner that is less costly and more timely than focus groups and personal interviews. Because of these advantages and another unique features, which Kozinets (2002) poses, the netnography methodology has been selected for Phase 1.

In the following sections, we aim to extract some handy rules for marketers based on the factors detected in Phase 1. In this regard, the rough set theory (RST) has been selected. RST can be regarded as a mathematical tool for the analysis of imperfect data. It has the ability to discover the rules governing phenomena and is able to view a series of conditions that result in consumer decision. The mathematical logic of RST accurately considers the uncertainty context of consumer content.

2. Literature survey with basic models

2.1 User-generated content and electronic WOM in tourism marketing

Social network sites (SNSs) have experienced outstanding expansion over the past decade, becoming essential channels for business communication and marketing, with an increasing number of companies that have created their own corporate pages on SNS (Herrero, Martin, and MarGarcia-De los Salmones, 2017). Users spread their experiences through SNSs. In business terminology, user-generated content (UGC), the process of uploading any self-created material through the internet by non-media is known as electronic WOM (eWOM). It works exactly like common WOM (Zanariah Bahtar and Muda, 2016).

In marketing survey, WOM is a substantial subject and a key player for promoting users or marketers. After using a product or service, many consumers share their experiences about the product or service with their friends and family (Albarq, 2014). With the

emergence of Web 2.0, UGC on networks is increasing every day. Online reviews provided by e-commerce websites as an important form of eWOM have played a significant part in helping patrons make decisions (Yan *et al.*, 2016). The entry of eWOM in the tourism industry is particularly well built. Intangibles such as tourism services cannot be evaluated before the using up experience. As a result, purchasing intangible products and services brings a higher risk and the clients are more dependent on the interpersonal influence of eWOM (Jalilvand and Samiei, 2012). With the development of information and communications technology (ICT), an increasing number of tourists obtain their travel information from the internet and handle transactions online (Miao, 2015). In other words, tourists discover new destinations or tourism products and act as organizers who share their opinions with others (Litvin *et al.*, 2008). According to some reports (TripAdvisor.com, 2016), the number of people who use online communities to review the comments of other tourists and choose a tourism destination includes more than hundreds of millions. The tourism industries in many countries follow the social medias as an essential device to promote their clients (Zeng and Gerritsen, 2014). Also, the offline meeting of the tourists at any destination apparently helps the users to make stronger relationships with related communities and it has a very strong motivation to obtain travel information (Chung and Buhalis, 2008). Pope *et al.* (2002) have found out that a large number of tourists make decision for selecting their destinations based on WOM sources. Moreover, reading of travel reviews helps to make travel planning more enjoyable to the tourists who feel more excitement about traveling (Parra-López *et al.*, 2011). On the other hand, Hennig-Thurau and Walsh (2004) have shown that consumers keep communicating their experiences with products or services to others via the web for a number of reasons. The desire for social interaction, economic incentives, concern for other consumers and potential to enhance self-worth are the primary factors that lead to eWOM behavior. Zhu and Lai (2009) have studied how online WOM influences tourism destination choice. They have found that the amount of information in WOM and consultations are positively related to the actual tourist reception. It means that a rational evaluation of WOM information regarding attributes (i.e. reliability, integrity, competence and quality assurance) is associated with a product, service or destination. According to the study of Mauri and Minazzi (2013), communication is limited to not only brands, products or services but also an organization, a destination, etc. Therefore, in the present study, the factors influencing the choice of destination is discovered by netnography, and in the next step, according to data discovered, the rules governing the behavior of tourists are examined using RST. Panda (2014) described the coordination mechanism of a socially responsible supply chain while any one firm is socially responsible for its stakeholders. Aviles-Gonzalez *et al.* (2016) searched recent reviews of tourism supply chain and studied tourism supply chain models considering different economics, marketing and optimization strategies. In the following, the basic concepts of netnography and RST are reviewed.

2.2 Rough set theory

RST proposed by Pawlak (1982) is a mathematical approach to data analysis. The basic idea of this method is based on classification of objects of interest into similar classes (clusters) containing objects, which are indiscernible blocks of knowledge about reality and are used to find out hidden patterns in data (Fazel Zarandi and Kazemi, 2008). This theory can be used to obtain some sets of attributes for classification and to evaluate the degree of the attributes of data based those are able to classify data (Kumar and Arasu, 2015). Samanta and Sarkar (2011) introduced fuzzy rough relation on a set and they proved that the collection of such relations is closed under different binary compositions.

Some basic notions of this theory are reviewed as follows.
In an information system called S :

$$S = (U, A, V, f) \quad (1)$$

Where, U is a non-empty finite set of objects, A is a non-empty finite set of attributes and V is the value set of A . The attributes include conditional attributes and decision attribute. Low approximation describes the scope of objects that certainly belong to the subset. Actually, the lower approximation of X is the largest community of all objects that clearly belong to the set X . The upper approximation includes the collection of objects that belong to the class of equivalence or equivalence classes of objects X which has at least one object in it. X series has always been placed between the two approximations that are divided into sections U District. Relations (2) and (3) represent lower and upper approximations, respectively, as follows:

$$\underline{apr}_p(X) = \cup \{x \in U: IND \subseteq X\} \quad (2)$$

$$\overline{apr}_p(X) = \cup \{x \in U: IND \cap X \neq \emptyset\} \quad (3)$$

Boundary region approximates the difference between upper and lower approximations (Figure 1). If X is empty, then set X is a crisp set, not a rough set:

$$BND_P(X) = \overline{apr}_p(X) - \underline{apr}_p(X) \quad (4)$$

If $S = (U, A, V, f)$ is an information system, Attribute A can be divided into two subsets which contains C and D as conditional and decision attributes, respectively. If Set A is completely dependent on C , it is displayed by $C \implies D$ and the degree of dependence of these two sets (k) is 1. But, sometimes it may not be entirely dependent. Then, it is a partial dependence ($0 < k < 1$) and its value is calculated as follows:

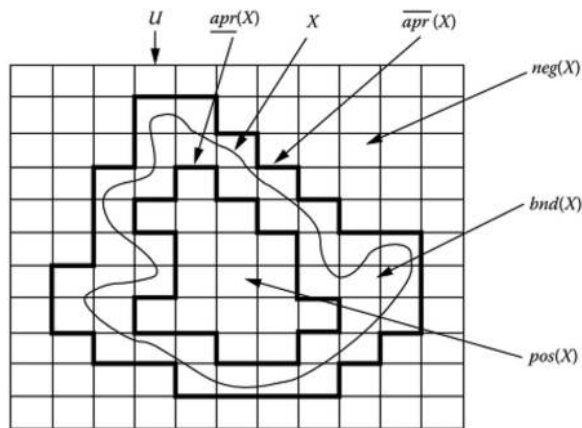


Figure 1.
Upper and lower approximations

$$k = \gamma(C, D) = \frac{|POS_C(D)|}{|U|} \quad (5) \quad \text{Application of rough set and netnography}$$

$$POS_C(D) = \cup_{x \in \frac{U}{D}} C(X) \quad (6)$$

Where, $POS_C(D)$ indicates the number of low approximation members in D and $x \in \frac{U}{D}$ represents a partition of a set of observations based on the decision attribute.

Another factor used to evaluate the accuracy is approximation accuracy. In an information system in the form $S = (U, A, V, f)$, the approximation accuracy is defined as follows:

$$\alpha_p|X| = \frac{|apr_p(X)|}{|\overline{apr_p}(X)|} \quad (7)$$

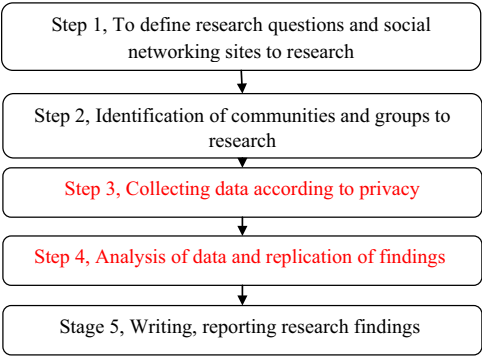
Here, $\alpha_p(X)$ indicates the approximate accuracy and $apr_p(X)$ and $\overline{apr_p}(X)$ also show the lower and upper approximations, respectively. This factor is a number between 0 and 1. If $\alpha_p(X) = 1$, the set X is considered as a crisp. In other words, if the number is less than 1, the set is a rough set.

Some attributes are ineffective or redundant, and summarizing information from these can be ignored. In this regard, Core and Reduct concepts are discussed in the following section. Reduct includes most small subsets of attributes that are seemingly independent and are able to decide equivalents to whole attributes. Core also includes the remaining common attributes of all Reducts.

2.3 Netnography

In the past two decades, the rise of the internet and the accompanying emergence of online communities and blogs have changed the way of communications of the consumers with each other. Therefore, the need of a systematic process to identify, select and analyze large volumes of online consumer conversations in less time and cost-effectively plays an important role in attracting customers (Bartl *et al.*, 2016). The advancement of internet has become popular to the consumers because of its openness of varieties of information which are observed in real life (Childress and Asamen, 1998). In this context, Kozinets (1997) has proposed netnography as an online ethnography conducted in a specific manner. It is an interpretive research method that adapts the traditional, in-person participant observation techniques of anthropology to the study of the interactions and experiences that manifest through digital communications. "Netnography," or ethnography on the internet, is a new qualitative research methodology that adapts ethnographic research techniques to the study of cultures and communities emerging through computer-mediated communications (Kozinets, 2002). Netnography is essentially costless compared with the traditional research which requires physical travel and costs of face-to-face field work, making more ambitious studies potentially more feasible (Xun and Reynolds, 2010). In this environment, store and track data are important and depend on the research questions and the available resources. The main steps of netnographic studies include data collection, analysis and interpretation. Figure 2 shows a schematic diagram of these steps briefly. The remarkable thing is that in

Figure 2.
Stages of
netnography



all stages of research ethics must be observed. [Bowler \(2010\)](#) has suggested accomplishing netnography following the research ethics:

- the researcher should fully disclose his or her presence, affiliations and intentions to online community members during any research;
- the researcher should ensure confidentiality and anonymity of informants; and
- the researcher should seek and incorporate feedback from members of the online community.

3. Methodology

This research considers a survey research according to the practical purpose involved in real-life problems. In this study, based on RST, there are two kinds of variable, namely, decision variable and conditional variable. Conditional attributes include the dimensions that affect customer behaviors in selecting touristic destination that is discovered through analysis of customer comments by using the method of netnography. The decision attribute measures tourists’ behavioral reaction according to their recent travel experiences, i.e. “if they travel to this destination again or if they recommend this destination to their friends or not.” The population of this research is the tourists who experienced at least once trip to one of these three cities: Isfahan, Shiraz or Tehran. This study consists of two phases: in the first phase of the study, the comments of people who have written about their travel experiences to one of the cities of Shiraz, Isfahan or Tehran on the TripAdvisor.com site was analyzed. As [Kozinets \(2010\)](#) said, the analysis process consists of coding, noting, abstracting and comparing, checking and refinement, generalizing and theorizing. Coding is the process of affixing codes or categories to data drawn from comments. Noting means that netnographics note important points in the margins of data to identify similar phrases, common sequences, relationships and distinct differences. Checking and refinement are to isolate, check and refine the understanding of patterns, processes, commonalities and differences. The nest stage of data analysis process is generalizing aims to explain the consistencies of netnographic data. The last step is theorizing, which involves inconsistent new theory in coordination with data and relevant body of knowledge. The review searching is continued until data saturation, i.e. when no new ideas or themes emerge, this is usually a good indication that enough data have been collected (Logan, 2015).

Then, based on discovered dimensions, a questionnaire is designed and distributed among tourists who have traveled to one of these three cities.

Therefore, the sample size based on the Cochran formula is 385. The approximate size of population has been assumed unlimited and the confidence level is 95 per cent:

$$n = \frac{\frac{z_{\frac{\alpha}{2}}^2 pq}{d^2}}{1 + \frac{1}{N} \left(\frac{z_{\frac{\alpha}{2}}^2 pq}{d^2} - 1 \right)} \cong 385.$$

To find the right population, the link of a questionnaire is randomly sent to 1,200 tourists (from 2,852 comments reviewed) who have expressed their travel experience in the form of comments on TripAdvisor.com. Among 1,200 tourists, 449 questionnaires are returned with the correct information. It should be noted that Cronbach's alpha coefficient is 0.882 for the overall questionnaire (beauty, 0.904; temperature condition, 0.878; cultural attraction, 0.796; safety, 0.895; welfare, 0.912; transportation quality, 0.863; local guides' quality, 0.876; costs, 0.932; local market quality, 0.842; dealing with passengers, 0.935; decision to visit again, 0.865) that shows the reliability of research tools. The validity of research has been reviewed and approved by ten experts. In this regard, content validity ratio (CVR) based on [Lawshe \(1975\)](#) is more than 0.62. After the distribution of the questionnaire, the consumer responses are collected in the form of linguistic variables.

Each variable is numerically scored from number 5 (strongly agree) to 1 (strongly disagree). Then according to the attached questionnaire, each dimension is determined on the basis of a number of questions. Therefore, the average number for each consumer determines the score of each dimension. The worth mentioning floating numbers are converted to the nearest integers.

Further, the rules are discover based on RST, and owing to the large number of objects (449 customer) the Rosetta software has been used. Here, [Figure 3](#) summarizes the process of the proposed research.

4. Results

As mentioned, in this study, first, the factors affecting consumer behavior in the tourism industry have been investigated. Based on this, the comments of tourists who have traveled to one of the cities Shiraz, Isfahan or Teheran and have shared their experiences on TripAdvisor.com are studied. Reviewing comments is continued until data saturation. So, 2,852 comments are reviewed, and the aforementioned ten main factors are taken into consideration.

In the next step, based on the mentioned dimensions, a questionnaire was designed and sent to the tourists who have experience traveling to the three cities; finally, 449 questionnaires with correct information were returned. [Table I](#) shows demographic variables of respondents based on the first section of the questionnaire.

According to the second section of the questionnaire, the average value of each dimension is calculated and shown in [Table II](#).

After measuring tourists' opinions, a logical connection between the collected data, based on RST, has been established. In the first step, according to the existing algorithms, Reduct and Core are calculated. The attributes that belongs to the Core include the most original and influential factors. The attributes of the Core have been determined as follows:

Core = {beauty, cultural attraction, safety, welfare, costs, dealing with passenger}.

Because of the large number of research data, 217 logical rules are extracted using RST and many of them have poor support factors. Therefore eight rules with the most frequency are selected regarding validation factors.

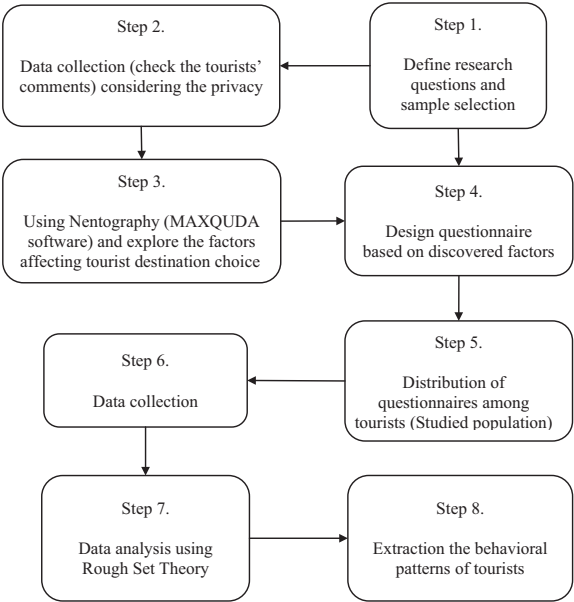


Figure 3.
Stages of research

Variables	<i>N</i>	(%)
<i>Gender</i>		
Male	231	51.45
Female	210	46.77
<i>Age</i>		
0-20	133	26.62
21-30	158	35.19
31-40	96	21.38
41-50	47	10.47
51-60	8	1.78
<i>Married status</i>		
Single	252	56.12
Married	190	42.32

Table I.
Demographic characteristics of study participants

According to the rules that have been extracted from the software, results can be summarized as follows:

- (1) If tourist destination has the following conditions, customers are truly satisfied and would recommend others to visit the region (decision attribute: very high):
 - “Beauty: very good,” “Cultural attractions: good”, “Security: good,” “Welfare state: good,” “Costs: very good,” “Dealing with passengers: good”.
 - “Beauty: good,” “Cultural attractions: very good”, “Security: good,” “Welfare state: good,” “Costs: very good,” “Dealing with passengers: good”.

If/then attributes	Respondent 1	Respondent 2	Respondent 3	...	Respondent 449
<i>If:</i>					
Beauty	5	4	3	...	4
Temperature condition	3	3	4	...	2
Cultural attraction	5	4	5	...	4
Safety	3	4	2	...	4
Welfare	3	4	4	...	3
Transportation quality	4	3	2	...	4
Local guides' quality	4	3	3	...	4
Costs	5	4	5	...	5
Local markets quality	3	2	3	...	4
Dealing with passengers	4	3	3	...	4
<i>Then:</i>					
Decide to visit again or recommend it to others	4	3	4	...	5

Table II.Average value of
data collected by
questionnaires

- (2) If tourist destination has the following conditions, customers are generally satisfied and would recommend to others to visit the region (decision attribute: high):
 - “Beauty: good,” “Cultural attractions: very good”, “Security: good,” “Welfare state: average,” “Costs: average,” “Dealing with passengers: good”.
 - “Beauty: good,” “Cultural attractions: good”, “Security: good,” “Welfare state: very good,” “Costs: average,” “Dealing with passengers: good”.
 - “Beauty: very good,” “Cultural attractions: good”, “Security: average,” “Welfare state: bad,” “Costs: good,” “Dealing with passengers: very good”.
- (3) If tourist destination has the following conditions, customers are dissatisfied and have little desire to recommend to others (decision attribute: average):
 - “Beauty: good,” “Cultural attractions: average”, “Security: good,” “Welfare state: average,” “Costs: average,” “Dealing with passengers: good”.
 - “Beauty: average,” “Cultural attractions: good”, “Security: bad,” “Welfare state: average,” “Costs: average,” “Dealing with passengers: bad”.
- (4) If tourist region has the following conditions, customers are truly dissatisfied and share their dissatisfaction with others (decision attribute: low):
 - “Beauty: average”, “Cultural attractions: good”, “Security: average,” “Welfare state: bad,” “Costs: bad,” “Dealing with passengers: good”.

5. Conclusions, implications, limitations and suggestions

In the decades after the Second World War, tourism growth was considerable in comparison with other part of the economy and it could gain better growth. Tourism industry is a profitable industry that creates jobs in the local and national area. Finally, economic development followed by social and cultural development causes growth in the national area. In this regard, it is necessary to present required strategies based on customer opinions.

In the meantime, with development and improvement of the internet, tourists can share their opinions in different ways. Thus, one of the methods of analyzing tourists' comments is

reviewing their comments on the website or virtual networks. Finally, by detailed reviewing of the comments and using RST, comments are analyzed and the rules governing tourists' behavior have been discovered.

Despite the natural resources and tourist attractions that make Iran one of the world's top ten countries in the tourism industry reported by World Tourism Organization, tourists prefer to choose other destinations. As a result, the share of tourism industry in terms of gross domestic product is negligible. Thus, Iran has eye-catching and cultural attractions potential of tourism. In this regard, the country needs greater recognition of the different areas that have the potential of nature tourism and detailed plan for the feasibility of the regions in terms of attracting tourists, creating different resorts and infrastructure facilities for them.

5.1 Conclusions

According to the results, welfare state is one of the most important for tourists when they want to choose a destination. If a traveler feels that the welfare conditions are not suitable, he/she will ignore that destination. Hotels that provide suitable conditions for accommodation of passengers have a very influential role in increasing the number of passengers who travel to a specific destination. In this regard, managers should provide facilities according to the international standards. Therefore, giving loans and creating opportunities for investors to build hotels can be very useful. Another determining factor that tourists face is high costs. Cost is one of the strategic factors caused by political, economic and cultural factors that can be managed with the right policies. To deal with passengers with an appropriate manner, managers' multiple cultural plan is required.

Future researchers can determine customers' behavior dimensions using other methods and compare them with the results of netnography. Researchers, by using different methods of multiple attribute decision making, can analyze the importance of these factors and rank them regarding another point of view. The next step can be to discover the rules or clustering the tourists by using another data mining method.

About rules that led to the result "very much," all attributes are approved with "good" or "very good" condition, and the attribute of cost is "very good" in both condition. About results that have led to "high" results, all three attributes, namely, eye-catching, cultural condition and dealing with passengers, have had "good" or "very good" condition and the other attributes also have relatively good condition. The result relating to average attributes indicates that the majority of the rules are in good or fair condition. In the end, the results relate to the good conditions of cultural attraction and dealing with passengers, whereas the other conditions are not suitable. Therefore, it causes tourists dissatisfaction. Finally, it can be concluded that beauty, cultural attractions and costs are the most important factors and deeply affect tourists' decision-making.

This paper helps managers analyze consumer behavior in destination marketing. In general, this research builds a framework to identify the most important factors in destination marketing based on UGC in social media and then extracts the rules of decision-making based on consumer behavior. Tourism sectors, especially in developing countries, need a proper strategy obtained from a micro-foundations perspective. The ultimate rules help managers to make better strategy and decide better.

Although the present research makes contributions to studies on tourism marketing, there are still limitations which might be taken into consideration for future studies. The achieved results are determined based on consumers (tourists) who are active on social media. These users have a common mind-set and personality then to make better decisions, and managers should also listen to the voice of offline tourism. Using the alternative

methods instead of a questionnaire could achieve more accurate results. Future researchers could resolve limitations mentioned before and apply this framework to other fields instead of tourism marketing.

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